

1. Introduction

The project aimed to design an agentic workflow capable of handling different types of user queries intelligently. Instead of relying on a single monolithic LLM, I implemented a multi-agent system using CrewAI and Groq API. Each agent had a specific responsibility, ensuring queries were routed correctly and answered with precision.

This case study explains how I created four custom agents, their roles, and how they collaborated to deliver structured, reliable, and fast responses.

2. Objective

The key objective was to:

- Build an intelligent query-handling system that could classify and route questions.
 - Integrate Groq API for high-speed, low-latency LLM inference.
 - Design modular agents, each focused on a specialized task.
 - Achieve better accuracy, maintainability, and scalability compared to a single-agent setup.
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3. Workflow Design

a) Validator Agent

The workflow begins with the Validator Agent, which acts as a gatekeeper.

- It checks whether the user's query is valid.
- It analyzes intent (e.g., general knowledge, diagram/table creation, or math reasoning).
- Based on this understanding, it routes the query to the appropriate specialized agent.

This ensures that irrelevant or poorly framed queries don't reach downstream agents.

b) General Q&A Agent

Once the validator identifies a query as a general information request, it is routed to the General Q&A Agent.

- This agent handles factual and explanatory answers.
- It is useful for queries like *“What is Reinforcement Learning?”* or *“Explain cloud computing.”*
- The focus here is accuracy and clarity.

This agent essentially acts as the knowledge expert in the system.

c) Diagram & Table Agent

If the query demands structured outputs, such as diagrams or tabular data, the Diagram & Table Agent takes over.

- It is capable of presenting comparisons, process flows, or structured data representation.
- Example queries include *“Show me a comparison between GPT-4 and LLaMA”* or *“Create a table of top AI frameworks.”*

This agent adds value by transforming raw text into more digestible visual or structured formats.

d) Math & Reasoning Agent

For queries that involve mathematical problem-solving or logical reasoning, the Math & Reasoning Agent is activated.

- It can solve equations, probability problems, and logical puzzles.
- It provides step-by-step reasoning to make solutions transparent.
- Example: A query like *“Solve $2x + 5 = 15$ ”* would be handled here.

This agent ensures that the system isn't limited to text explanations but can also perform analytical computations.

4. Execution Flow

1. A user submits a query.
2. The Validator Agent analyzes it and determines intent.
3. The query is routed to either the General Q&A, Diagram & Table, or Math & Reasoning Agent.
4. The chosen agent processes the query using Groq-powered inference.
5. The system compiles the response and returns it to the user.

This structured flow ensures queries are handled with specialized precision instead of generic responses.

5. Results & Impact

The agentic workflow delivered several benefits:

- **Accuracy:** Specialized agents improved the relevance and correctness of responses.
 - **Efficiency:** The Groq API ensured rapid inference, reducing latency.
 - **Scalability:** The modular design made it easy to add future agents (e.g., a RAG-based knowledge agent).
 - **User Experience:** The system could seamlessly shift between explanation, structured data, and problem-solving without confusion.
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6. Conclusion

This project demonstrated the power of agentic workflows in handling diverse user queries. By combining CrewAI's orchestration with Groq's speed, I created a robust, modular, and scalable system.

Instead of depending on a single agent, the use of four specialized agents — Validator, General Q&A, Diagram & Table, and Math & Reasoning — ensured that each query was validated, understood, and answered with the right expertise.

The result is a workflow that feels intelligent, fast, and domain-aware, making it suitable for real-world use cases like educational platforms, AI-powered assistants, and enterprise query systems.